H1N1 Outbreak Model Report

11-Compartment ODE Model with a 9-Day Infectious Period

Project Report

Numerical Methods / Mathematical Modeling

Topic:	H1N1 outbreak dynamics in a closed population
Population size:	$N = 1500$
Infectious period:	9 days
Initial condition:	1 infected, 1499 susceptible, 0 recovered
Key calibration target:	total infected after 7 days equals 10

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1 Problem Statement

This project studies the spread of an H1N1-like virus through a closed population of size $N = 1500$. At time $t = 0$, exactly one person is infected and the remaining 1499 people are susceptible. Once a person becomes infected, that person remains infectious for exactly 9 days. After the 9-day infectious period, the person moves into a recovered class and is assumed to remain immune for the rest of the outbreak.

The goal of the project is to build a compartmental differential equation model that reflects this 9-day infectious period, determine the unknown transmission rate β, numerically solve the system, and interpret the resulting outbreak behavior. In particular, the model must satisfy the condition that after 7 days there are 10 total infected individuals.

The report is organized according to the required project components: the problem statement, background and model assumptions, methodology, mathematical work, numerical results, conclusions, and references.

2 Background Material and Model Setup

2.1 Why an 11-compartment model?

A standard SIR model contains only three compartments: susceptible, infected, and recovered. In this project, however, the infectious period is known more precisely: infected individuals remain infectious for exactly 9 days. To represent that structure, the infected population is split into 9 separate stages,

$$I_1(t), I_2(t), \ldots, I_9(t),$$

where $I_j(t)$ denotes the number of people on day j of infection at time t.

This leads to a total of 11 compartments:

$$S(t),\ I_1(t), I_2(t), \ldots, I_9(t),\ R(t).$$

This stage-structured formulation is more realistic than using a single infected compartment because it explicitly tracks how individuals progress through infection before recovering.

2.2 Model assumptions

The model is based on the following assumptions:

1. The population is closed, so there are no births, deaths, immigration, or emigration.
2. Susceptible individuals become infected through contact with currently infected individuals.

3. Transmission follows mass-action incidence.
4. Each infected individual spends exactly one day in each infected class $I_1, I_2, \ldots, I_9$.
5. After completing the ninth day of infection, the individual enters the recovered class.
6. Recovered individuals remain immune throughout the outbreak.

The total infected population at time t is therefore

$$I(t) = \sum_{j=1}^{9} I_j(t).$$

The force of infection uses the usual mass-action term

$$\frac{\beta}{N} S(t) I(t),$$

where β is the transmission parameter to be determined.

2.3 Runge-Kutta method as numerical background

Because the outbreak model is a coupled nonlinear system, it does not have a simple closed-form solution. A standard numerical background method for first-order ODE systems is the Runge-Kutta method.

For a general system

$$\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}), \qquad \mathbf{y}(t_n) = \mathbf{y}_n,$$

the classical fourth-order Runge-Kutta method (RK4) advances the solution from t_n to $t_{n+1} = t_n + h$ by computing four intermediate slopes:

$$\begin{aligned}
k_1 &= \mathbf{f}(t_n, \mathbf{y}_n), \\
k_2 &= \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} k_1\right), \\
k_3 &= \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} k_2\right), \\
k_4 &= \mathbf{f}(t_n + h, \mathbf{y}_n + h k_3).
\end{aligned}$$

The next numerical value is then approximated by

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6} \left(k_1 + 2k_2 + 2k_3 + k_4\right).$$

In this project, the unknown function is the 11-component state vector

$$\mathbf{y}(t) = \left(S(t), I_1(t), I_2(t), \ldots, I_9(t), R(t)\right)^T.$$

So a Runge-Kutta solver advances all 11 compartments simultaneously by evaluating the right-hand side of the model several times within each time step.

In the actual computation, the Python solver `solve_ivp` was used with method `RK45`. This is an adaptive Runge-Kutta method from the same family, so it automatically adjusts the step size while using the same basic idea of combining several slope estimates to obtain an accurate numerical solution.

3 Mathematical Formulation of the Model

3.1 Differential equation system

Using the assumptions above, the outbreak is modeled by the following system of ordinary differential equations:

$$
\begin{aligned}
\frac{dS}{dt} &= -\frac{\beta}{1500} S(I_1 + I_2 + \cdots + I_9), \\
\frac{dI_1}{dt} &= \frac{\beta}{1500} S(I_1 + I_2 + \cdots + I_9) - I_1, \\
\frac{dI_2}{dt} &= I_1 - I_2, \\
\frac{dI_3}{dt} &= I_2 - I_3, \\
\frac{dI_4}{dt} &= I_3 - I_4, \\
\frac{dI_5}{dt} &= I_4 - I_5, \\
\frac{dI_6}{dt} &= I_5 - I_6, \\
\frac{dI_7}{dt} &= I_6 - I_7, \\
\frac{dI_8}{dt} &= I_7 - I_8, \\
\frac{dI_9}{dt} &= I_8 - I_9, \\
\frac{dR}{dt} &= I_9.
\end{aligned}
$$

The initial conditions are

$$
S(0) = 1499, \qquad I_1(0) = 1, \qquad I_2(0) = \cdots = I_9(0) = 0, \qquad R(0) = 0.
$$

3.2 Conservation of population

A useful check on the model is that the total population remains constant. If we add all 11 equations together, then every internal transfer cancels and we obtain

$$\frac{d}{dt}\left(S + I_1 + I_2 + \cdots + I_9 + R\right) = 0.$$

Therefore,

$$S(t) + \sum_{j=1}^{9} I_j(t) + R(t) = 1500$$

for all time t. This confirms that the model conserves population.

3.3 Substituting the numerical values into the equations

From the numerical calibration step, the transmission parameter is found to be

$$\beta \approx 0.339976 \text{ day}^{-1}.$$

Since the population size is $N = 1500$, the transmission coefficient in the mass-action term becomes

$$\frac{\beta}{N} = \frac{0.339976}{1500} \approx 0.0002266507.$$

Thus the first two equations may be written numerically as

$$\begin{aligned}
\frac{dS}{dt} &= -0.0002266507\, S\big(I_1 + I_2 + \cdots + I_9\big),\\
\frac{dI_1}{dt} &= 0.0002266507\, S\big(I_1 + I_2 + \cdots + I_9\big) - I_1.
\end{aligned}$$

The remaining equations are unchanged:

$$\begin{aligned}
\frac{dI_2}{dt} &= I_1 - I_2, & \frac{dI_3}{dt} &= I_2 - I_3, & \frac{dI_4}{dt} &= I_3 - I_4,\\
\frac{dI_5}{dt} &= I_4 - I_5, & \frac{dI_6}{dt} &= I_5 - I_6, & \frac{dI_7}{dt} &= I_6 - I_7,\\
\frac{dI_8}{dt} &= I_7 - I_8, & \frac{dI_9}{dt} &= I_8 - I_9, & \frac{dR}{dt} &= I_9.
\end{aligned}$$

At the initial time $t = 0$, these substitutions give

$$I(0) = I_1(0) + \cdots + I_9(0) = 1,$$

so the initial infection rate is

$$\frac{\beta}{1500} S(0)I(0) = \frac{0.339976}{1500}(1499)(1) \approx 0.33975.$$

Therefore,

$$\left.\frac{dS}{dt}\right|_{t=0} \approx -0.33975, \qquad \left.\frac{dI_1}{dt}\right|_{t=0} \approx 0.33975 - 1 = -0.66025.$$

This means that initially the susceptible class begins decreasing slowly, while the first infected class decreases because individuals are simultaneously moving from day 1 to day 2 of infection.

4 Methodology

4.1 Why numerical methods are required

The model is nonlinear because the infection term involves the product $S(t)I(t)$. Because of this nonlinearity and because the infected population is split across 9 coupled stages, there is no simple closed-form solution for the full system. For that reason, the system must be solved numerically. The Runge-Kutta background summarized in the previous section provides the numerical framework used to advance the solution in time.

4.2 Calibration of the transmission parameter

The unknown parameter β is chosen so that the model satisfies the project condition

$$I(7) = \sum_{j=1}^{9} I_j(7) = 10.$$

To do this, we define a function

$$f(\beta) = I(7;\beta) - 10,$$

where $I(7;\beta)$ is the total infected population at day 7 obtained by numerically solving the system with a trial value of β. The desired value of β is the root of this equation:

$$f(\beta) = 0.$$

Using numerical root-finding together with repeated ODE simulation, we obtain

$$\beta \approx 0.339976.$$

4.3 Numerical solution procedure

The main computational steps were:

1. Define the 11-compartment ODE system.
2. Set the initial conditions $S(0) = 1499$, $I_1(0) = 1$, $I_2(0) = \cdots = I_9(0) = 0$, and $R(0) = 0$.

3. Use a numerical ODE solver based on Runge-Kutta methods to simulate the system over time.
4. Use a root-finding algorithm to adjust β until the condition $I(7) = 10$ is met.
5. Plot the total susceptible, infected, and recovered populations.
6. Plot each infected stage $I_1, \ldots, I_9$ separately to visualize progression through the 9-day infectious period.

4.4 Python implementation used in the methodology

The following Python workflow was used in the numerical part of the project:

- `numpy` for numerical arrays,
- `scipy.integrate.solve_ivp` for solving the ODE system,
- `scipy.optimize.root_scalar` for determining β,
- `matplotlib` for producing the figures.

A short implementation excerpt is shown below.

Listing 1: Core numerical workflow used in the methodology.

```
import numpy as np
from scipy.integrate import solve_ivp
from scipy.optimize import root_scalar

N = 1500


def h1n1_system(t, y, beta):
    S = y[0]
    I = y[1:10]
    R = y[10]
    Itot = np.sum(I)

    dydt = np.zeros(11)
    dydt[0] = -(beta / N) * S * Itot
    dydt[1] = (beta / N) * S * Itot - I[0]
    for j in range(1, 9):
        dydt[j + 1] = I[j - 1] - I[j]
    dydt[10] = I[8]
    return dydt
```

```
def total_infected_day7(beta):
    y0 = [1499, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
    sol = solve_ivp(
        lambda t, y: h1n1_system(t, y, beta),
        [0, 7], y0, t_eval=[7], method="RK45"
    )
    return np.sum(sol.y[1:10, -1])


def objective(beta):
    return total_infected_day7(beta) - 10

beta_star = root_scalar(objective, bracket=[0.1, 1.0], method="brentq").
   root
```

The complete solver code is included in Appendix A.

5 Results

The numerical solution gives the following main results:

Quantity	**Approximate value**
Transmission parameter β	0.339976
Peak total infected population	787.77 people
Time at peak infection	27.1 days
Final susceptible population	83.32 people
Final recovered population	1416.68 people

These values show that the outbreak spreads through most of the population. Only about 83 people remain uninfected by the end of the simulation, while more than 1416 people eventually move into the recovered class.

6 Interpretation of the Graphs

6.1 Figure 1: Susceptible, total infected, and recovered populations

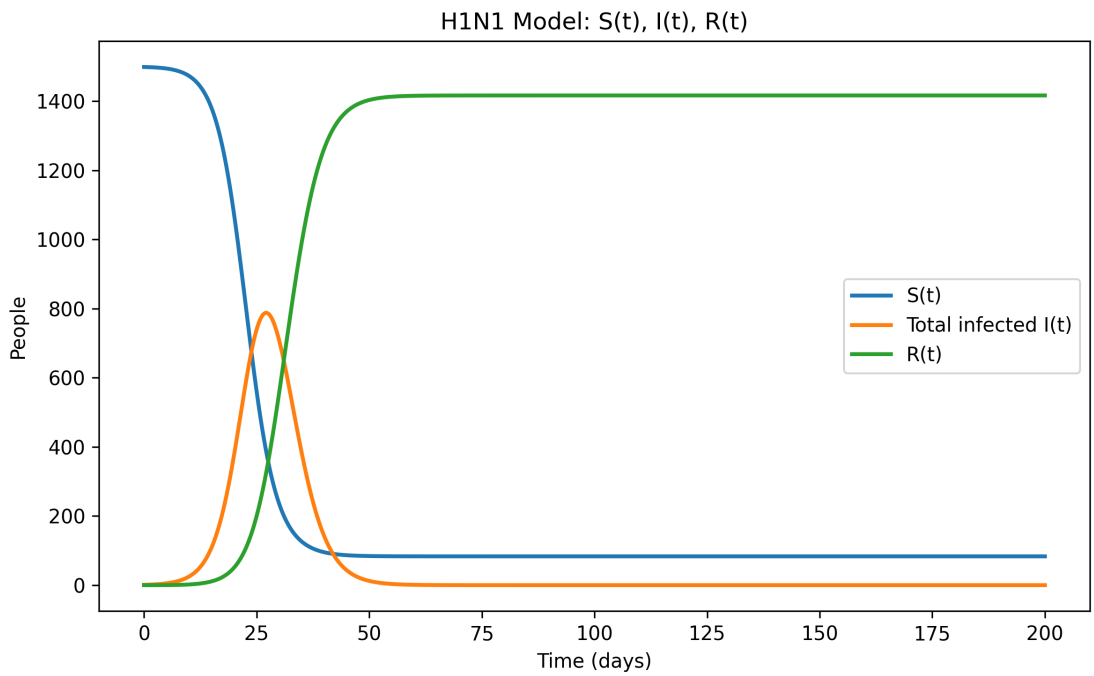

Figure 1: Susceptible, total infected, and recovered populations over time.

Figure 1 shows the overall epidemic trajectory. At the beginning of the outbreak, nearly the entire population is susceptible, so the disease has a large pool of people to infect. This is why the infected curve rises rapidly after the first few days.

The blue curve, $S(t)$, starts near 1500 and falls sharply once transmission accelerates. This drop indicates that susceptible individuals are being converted into newly infected individuals at a high rate.

The orange curve, which represents the total infected population

$$I(t) = \sum_{j=1}^{9} I_j(t),$$

first increases, then reaches a maximum of about 788 people around day 27, and then decreases. The decrease after the peak occurs because fewer susceptible individuals remain available to infect, while infected individuals continue progressing through the 9 infectious stages and then recover.

The green curve, $R(t)$, increases steadily over time because recovered individuals accumulate and do not leave the recovered class. By the end of the outbreak, the recovered population levels off near 1417, showing that most of the population experienced infection at some point.

Overall, Figure 1 shows the standard pattern of a major epidemic wave: a large susceptible pool

at the start, a rapid rise to peak infection, and eventual decline as the susceptible population is depleted.

6.2 Figure 2: Infected compartments by day of infection

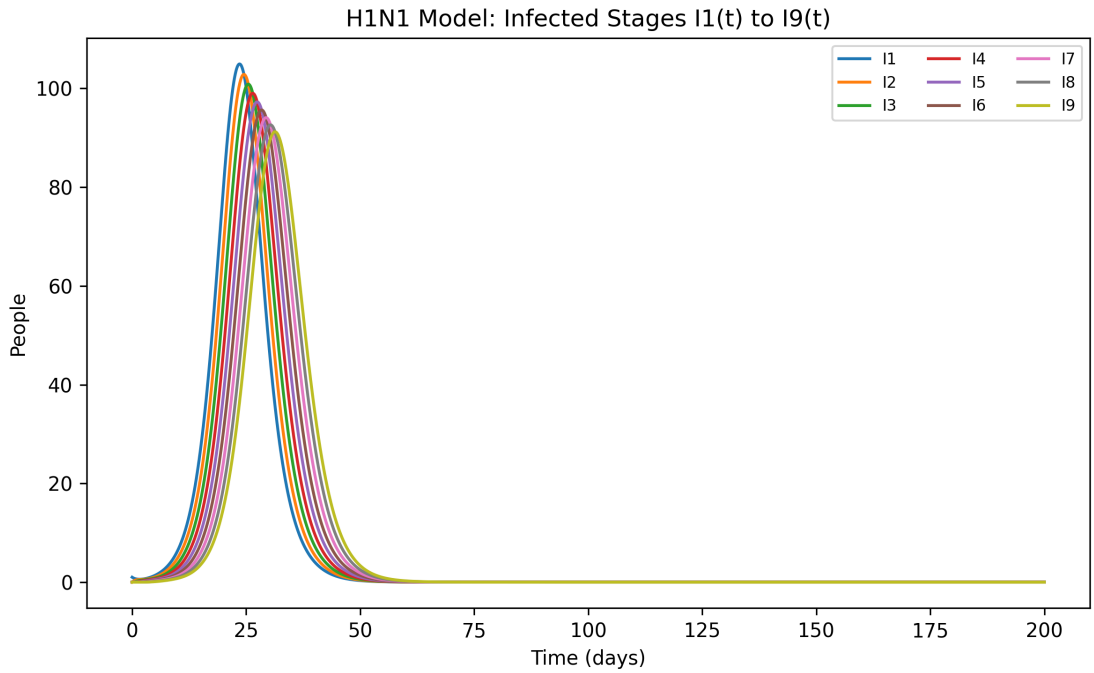

Figure 2: Infected compartments $I_1(t)$ through $I_9(t)$.

Figure 2 breaks the total infected population into the nine infection stages. Each curve corresponds to one day of infection: I_1 is day 1, I_2 is day 2, and so on through I_9.

The curves have similar bell-shaped profiles, but they are shifted in time. The I_1 curve peaks first because newly infected individuals enter the first stage immediately. Then, as those individuals progress through the infectious period, the peaks of $I_2, I_3, \ldots, I_9$ occur one after another. This staggered pattern is exactly what we expect from a model in which individuals remain infectious for a fixed 9-day period and move sequentially through the stages.

The graph also shows that there is a smooth flow of people through the infected compartments. No compartment behaves erratically, which supports the consistency of the numerical solution. The similar peak heights indicate that during the most intense part of the outbreak, large numbers of individuals are present in every stage of infection.

In short, Figure 2 confirms that the stage-structured model is working as intended: infection enters through I_1, progresses from one day of infection to the next, and finally exits to the recovered class after the ninth stage.

7 Discussion

The computed value $\beta \approx 0.339976$ produces an outbreak that satisfies the requirement that 10 total individuals are infected after one week. Once that calibration condition is imposed, the model predicts a large epidemic wave in the closed population.

One reason the outbreak becomes so large is that the initial susceptible population is extremely high relative to the number initially infected. Since almost everyone is susceptible at the beginning, the transmission term

$$\frac{\beta}{1500} SI$$

is initially able to generate many new infections as soon as the infected population begins growing.

The model also demonstrates the value of using multiple infected compartments. If only one infected compartment were used, the model would not explicitly track how long each person has been infected. Here, however, the 9 infected classes give a clearer biological interpretation: every person spends one day in each infection stage before recovery. That structure makes the progression of the disease easier to interpret and better reflects the problem statement.

8 Conclusion

In this project, we constructed and solved an 11-compartment epidemiological model for an H1N1-like outbreak in a closed population of 1500 individuals. The model was designed to reflect a 9-day infectious period by dividing infected individuals into nine separate stages.

Because the resulting system is nonlinear, numerical methods were required. We used an ODE solver together with a root-finding procedure to determine the transmission parameter β so that the model satisfied the condition $I(7) = 10$.

The final calibrated model gave

$$\beta \approx 0.339976,$$

and predicted a substantial outbreak with a peak infected population of approximately 788 people around day 27. By the end of the outbreak, only about 83 susceptible individuals remain, while about 1417 individuals have recovered.

Overall, the project shows how compartmental differential equation models can be used to represent infectious disease spread, incorporate biological structure such as a fixed infectious period, and generate meaningful quantitative predictions through numerical simulation.

9 References